



Original Article/Research

Individual factors that affect laypeople's understanding of definitions of medical jargon

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ABSTRACT

Objective: Patients have difficulty understanding medical jargon in electronic health record (EHR) notes. Lay definitions can improve patient comprehension, which is the goal of the NoteAid project. We assess whether the NoteAid definitions are understandable to laypeople and whether understandability differs with respect to layperson characteristics.

Methods: Definitions for jargon terms were written for laypersons with a 4th-to-7th-grade reading level. 300 definitions were randomly sampled from a corpus of approximately 30,000 definitions. 280 laypeople (crowdsource workers) were recruited; each layperson rated the understandability of 20 definitions. Understandability was rated on a 5-point scale. Using a generalized estimating equation model (GEE) we analyzed the relationship between understandability and age, sex, race/ethnicity, education level, native language, health literacy, and definition writer.

Results: Overall, 81.1 % (95 % CI: 76.5–85.7 %) of the laypeople reported that the definitions were understandable. Males were less likely to report understanding the definitions than females (OR: 0.73, 95 % CI: 0.63–0.84). Asians, Hispanics, and those who marked their race/ethnicity as “other” were more likely to report understanding the definitions than whites (Asians: OR: 1.43, 95 % CI: 1.17–1.73; Hispanics: OR: 1.86, 95 % CI: 1.33–2.59; Other: OR: 2.48, 95 % CI: 1.65–3.74). Laypeople whose native language was not English were less likely to report understanding the definitions (OR: 0.51, 95 % CI: 0.36–0.74). Laypeople with lower health literacy were less likely to report understanding definitions (health literacy score 3: OR: 0.51, 95 % CI: 0.43–0.62; health literacy score 4: OR: 0.40, 95 % CI: 0.29–0.55).

Conclusion: Understandability of definitions among laypeople was high. There were statistically significant race/ethnic differences in self-reported understandability, even after controlling for multiple demographics.

Public interest summary: We conducted a study to ensure that definitions written for the NoteAid EHR jargon identification tool are understandable. We recruited a diverse group of crowdsource workers and found that overall, the definitions were understandable, but understanding levels varied based on several demographic characteristics.

Introduction

Recent federal regulations mandate that healthcare providers make

patient medical records, including provider notes, available to patients via their electronic health records (EHRs) [1]. Therefore, approximately 200 million American adults can now read their own EHR notes written

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by physicians and other medical professionals [2].

However, EHR notes are written by medical professionals for other medical professionals. Therefore, the notes contain complex medical jargon, abbreviations, and acronyms that can be difficult for patients to understand. For example, the following excerpt is from an inpatient EHR that describes an upcoming surgery:

“From a GI standpoint, we recommend to proceed with bariatric surgery. However, he will need to continue daily PPI administration to maximize acid reduction.”

Patient access to EHR notes has benefits. For example, certain subsets of patients are highly motivated to monitor their health. These patients are referred to as “high activation” patients and are typically younger and have private health insurance, higher income, and higher education levels [3,4]. They typically read their EHR notes more frequently, follow-up with their care providers with questions, and ensure that they are engaged with their care. Research has shown that increased patient activation is associated with a number of improved health outcomes, including more healthy behaviors and more use of preventative screening tests [5]. High-activation patients also have higher patient satisfaction scores and better clinical indicators [5–7].

Access to EHR records is a promising venue for patients who want to increase their engagement with their own care. However, recent work has shown that patient engagement with their own EHRs is not consistent across demographic groups. Patients who are a minority, live in rural areas, are older, or are not native-English speakers are less likely to access their EHRs [7]. These groups are also traditionally linked to lower health literacy in general [7–10]. Health literacy has been defined as “a set of skills that people need to function effectively in the health care environment” [11], “people’s knowledge, motivation and competences to access, understand, appraise, and apply health information in order to make judgments and take decisions in everyday life concerning healthcare, disease prevention and health promotion to maintain or improve quality of life during the life course,” [12] and “the ability to exercise critical judgement of health information and resources, as well as the ability to interact and express personal and societal needs for promoting health.” [13]

It is estimated that 80 million Americans have limited health literacy [7,11,14]. These patients are less able to understand jargon and acronyms associated with their EHR notes. Health literacy and patient activation are positively correlated with each other, and both are independently correlated with health outcomes such as overall health, depression, and anxiety [15]. Low health literacy is costly, both in terms of health outcomes and healthcare expenses. Low health literacy is associated with poor communication between patients and physicians with negative health outcomes from pediatrics to diabetes [11]. Issues traced back to low activation account for 3–5 % of healthcare expenses [11,16]. For example, low-activation patients with asthma had costs that were 21 % higher than high-activation patients with asthma, after controlling for demographic characteristics and healthcare risk [16]. Annual spending on healthcare in the US is \$4.3 trillion as of 2021 [17]. Assuming a conservative estimate of 3 % for expenses related to low health literacy [18], this amounts to a \$129 billion problem [19].

We therefore have developed NoteAid [20], a system that uses natural language processing techniques to automatically identify and rank jargon in EHR notes (Fig. 1). For identified jargon, NoteAid then retrieves and presents a lay language definition for the jargon term in the web interface. Prior work has shown that NoteAid can improve patient eHealth literacy [21,22].

As a part of the NoteAid project, we have written over 30,000 human-generated lay definitions for medical jargon. Our prior work has shown that NoteAid is effective for improving EHR note comprehension [21,23]. A critical question for resources such as the NoteAid dictionary is with regards to the understandability of the written definitions. Prior work has shown that a majority of patient educational materials are written at an 8th-grade level or above [24]. However, The Joint

Commission recommends that educational materials for patients should be written at a 5th-grade reading level or lower [25]. In order for the resource to be useful for patients, we must ensure that guidelines for writing definitions are appropriate and lead to understandable definitions. What’s more, prior work showing discrepancies between readability and understandability for health texts suggest the need for human-level assessment of definitions beyond automated metrics [26].

Our research objectives of this study were to assess a) whether the definitions written by the NoteAid definition writers are understandable to laypeople and b) whether understandability differs with respect to layperson characteristics.

Methods

To address our research objectives, we conducted an observational study, using crowdsourced workers from Amazon Mechanical Turk (MTurk) as proxies for patients [27]. This study was conducted with approval from the Institutional Review Board at the University of Massachusetts Lowell.

NoteAid dictionary development

The NoteAid dictionary contains human-generated definitions for medical jargon and abbreviations [28]. Jargon terms were extracted from the clinical notes of de-identified EHRs (emergency department notes, progress notes, consult notes, operative reports, radiology reports, pathology reports, and discharge summaries). Potential jargon terms were identified with natural language processing techniques [28]. Specifically, EHR notes were tokenized using the Natural Language Toolkit.¹ A list of tokens was compiled, stop words excluded, and filtered against the Unified Medical Language System Metathesaurus.² The resulting tokens were presented to definition writers in order of their frequency in the EHR corpus, along with source sentences. Final selection of jargon and writing of definitions were done by experienced lay and clinician definition writers.

Definition writers were instructed to write definitions for laypersons with a 4th-to-7th-grade reading level and were shown the definition’s Flesch-Kincaid Grade Level (FKGL) score [29,30] while it was being written. Definition scoring above 7 could be used at the definition writer’s discretion. The FKGL score is defined as:

$$0.39 \left(\frac{\text{total words}}{\text{total sentences}} \right) + 11.8 \left(\frac{\text{total syllables}}{\text{total words}} \right) - 15.59$$

We used the dictionary-based tool Pyphen to count the number of syllables.³ Using the FKGL is consistent with prior work examining readability of online text [31–34]. Incomplete sentences were treated as complete sentences when calculating the FKGL score, which may result in lower scores. Definition writers followed a set of guidelines to help them select jargon and write definitions (see the supplemental material for excerpt from the guidelines). The guidelines were developed using existing materials from the literature and included plain-language guidelines for general US government documents [35] as well as guidelines for healthcare-specific documents [36,37]. Guidelines include information on jargon and selection, word choice, sentence construction, extent of detail, and information sources. For example, the guidelines say that to define a treatment, the definition should include the type of treatment and its purpose. For instance, “capsular release” should be defined as “surgery to allow a joint to move.”

¹ <https://www.nltk.org>

² https://www.nlm.nih.gov/research/umls/knowledge_sources/metathesaurus/index.html

³ <https://github.com/Kozee/Pyphen/tree/main>

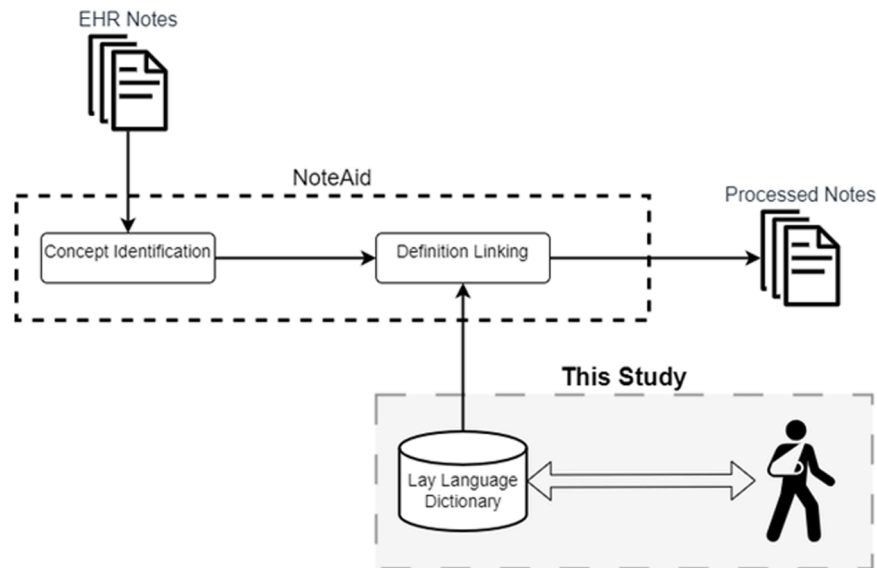


Fig. 1. Overview of the NoteAid system. These definitions are fed into NoteAid and then presented to the user with the processed EHR notes. Therefore, it is important that the definitions in the database are understandable. In this study, we use crowdsource workers to assess the understandability of the Lay Language Dictionary definitions.

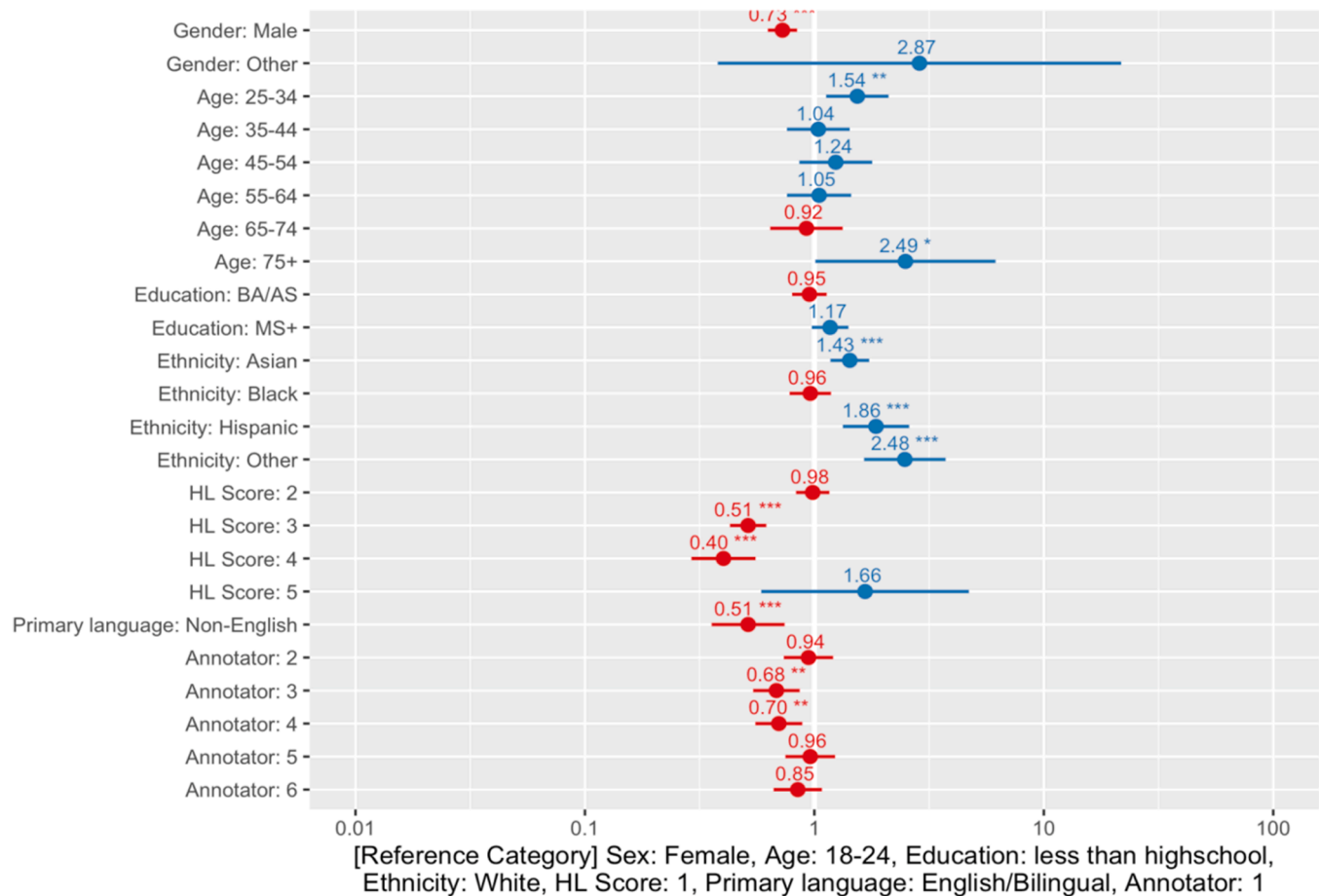


Fig. 2. GEE odds ratios for understanding definitions compared to demographic reference groups. Those with odds ratios significantly different from one are noted in the plot: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Definition and selection of terms

From the approximately 30,000 definitions in the NoteAid corpus, 300 were sampled for study. For each of six definition writers, we randomly selected definitions and applied the following inclusion criteria until we had sampled 50 definitions from each writer (300 in total):

- Date: The definition was written between 6/1/2020 and 10/31/2020.
- Length: The source EHR sentence was between 20 characters and 700 characters.
- Substance: The definition text did not consist solely of an acronym expansion, spelling correction or reference to another term with the same definition, without any actual definition provided.
- Relevance: The term being defined was jargon, as judged by two members of the research team.
- Presence of jargon in the definition: The NoteAid system was designed to allow definition writers to include jargon in their definitions if doing so would substantially shorten or simplify the definition. In those cases, a second definition for the “embedded” jargon would then be supplied to the user. In this study, such definitions were discarded to avoid MTurk workers rating two definitions simultaneously (Fig. 2).
- History: The definition had not been revised by another definition writer, which would make attribution to a specific definition writer difficult.

The mean length of sampled definitions was 9.3 words (range 1–35) and 1.1 sentences (range 1–3). The range of topic areas is very broad, encompassing anything that might occur in an EHR note, for example, abbreviations, drugs, pathogens, symptoms, procedures, and lab tests. The full list of sampled definitions is included in the supplemental materials.

Subjects

Subjects were recruited from Amazon Mechanical Turk (MTurk). MTurk allows for efficient hiring of workers for discrete tasks [38–40]. We recruited United States–based, English-speaking MTurk workers aged 18 and over and excluded subjects in healthcare professions. We excluded MTurk workers who received below a 95 % approval rating for their previous work on the MTurk platform. In addition, we embedded attention checks into the task to exclude MTurk workers who were not paying attention. Individuals who failed the attention checks were asked to begin the task again. 26 MTurk workers did not complete the study, either due to dropping out or failed attention checks; their results were not included in the analyses. Additional details on the attention check design are included in the supplemental materials.

Sample and measures

Our sample consisted of 280 MTurk workers. Prior work has also shown that understandability and health literacy vary across different demographic groups [41]. We analyzed whether NoteAid definitions performed equally well for different demographic subgroups (age, sex, race/ethnicity, and education) and for other factors that would be expected to affect definition understanding (native language, health literacy, and definition writer).

We oversampled certain groups because the MTurk population is skewed, for example, towards young, white individuals with more higher education [42,43]. To do this, we created 18 subgroups and divided our desired demographics accordingly. We then worked with a 3rd-party to assist with Turker recruitment to ensure that we were able to recruit MTurk workers that fit the groups we were targeting. Recruitment continued until each group was filled. Additional details

are included in the supplemental materials.

Health literacy was measured using the Single Item Literacy Screener [44], in which subjects were asked the following question: “How often do you need someone’s help when reading instructions given by your doctor or pharmacist?” Response choices ranged from “never” to “always.”

Subjects were shown a jargon term, the definition from the NoteAid dictionary, and the sentence in the EHR note where the jargon term appeared. Each MTurk worker was given 20 definitions to evaluate. Each definition was rated by at least 18 MTurk workers. MTurk workers were recruited in batches, and the definitions were allocated to the MTurk workers in sequence [45]. MTurk workers were asked to rate the understandability of each definition on the following 5-point scale:

1. Not understandable
2. Slightly understandable
3. Somewhat understandable
4. Understandable
5. Completely understandable

Statistical analysis

From the 5-point understandability rating, we constructed a binary outcome measure: definitions that were rated 4 or 5 were designated as *understandable* and definitions that were rated 1, 2, or 3 were designated as *not understandable*. Our dependent variable, *understandability*, was therefore the mean of this binarized variable (range 0 to 1) across all of the definitions rated.

We analyzed the relationship between self-reported understandability and MTurk worker characteristics. As described above, each subject was shown 20 such definitions. The observations were not independent of each other, so we fit a generalized estimating equation model (GEE) to adjust for demographic characteristics, definition writer,

Table 1
MTurk worker demographic characteristics.

Characteristic	N = 280
Gender	
Female	152 (54 %)
Male	127 (45 %)
Other	1 (0.4 %)
Age	
18–24	17 (6.1 %)
25–34	77 (27.5 %)
35–44	62 (22.1 %)
45–54	31 (11.1 %)
55–64	67 (23.9 %)
65–74	23 (8.2 %)
75 and over	3 (1.1 %)
Education	
High school diploma or less	93 (33 %)
Bachelor’s or associate’s degree	95 (34 %)
Master’s degree or above	92 (33 %)
Race	
White	142 (51 %)
Asian	63 (23 %)
Black	44 (16 %)
Hispanic	16 (5.7 %)
Other	15 (5.0 %)
Native language	
English or bilingual	271 (97 %)
Non-English	9 (3.2 %)
Health literacy score*	
1 - Never	136 (49 %)
2 - Rarely	83 (30 %)
3 - Sometimes	47 (17 %)
4 - Often	12 (4.3 %)
5 - Always	2 (0.7 %)

* Response to the question: “How often do you need someone’s help when reading instructions given by your doctor or pharmacist?”.

Table 2
Percentage understandable by MTurk worker characteristic.

MTurk worker characteristic	n	% understandable*	95 % CI	
			LCL	UCL
Overall	280	81.1 %	76.5 %	85.7 %
Gender				
Female	152	82.4 %	76.3 %	88.5 %
Male	127	79.4 %	72.4 %	86.5 %
Other	1	95.0 %	52.3 %	100 %
Age				
18–34	17	76.8 %	56.7 %	96.8 %
25–34	77	84.3 %	76.2 %	92.4 %
35–44	62	78.5 %	68.3 %	88.8 %
45–54	31	83.7 %	70.7 %	96.7 %
55–64	67	80.7 %	71.2 %	90.1 %
65–74	23	77.2 %	60.0 %	94.3 %
75 or older	3	90.0 %	56.1 %	100 %
Education				
High school diploma or less	93	80.1 %	72.0 %	88.2 %
Bachelor's or associate's degree	95	79.3 %	71.2 %	87.5 %
Master's degree or above	92	84.0 %	76.5 %	91.5 %
Race				
White	142	79.9 %	73.3 %	86.5 %
Asian	63	81.8 %	72.3 %	91.3 %
Black	44	79.2 %	67.2 %	91.2 %
Hispanic	16	85.6 %	68.4 %	100 %
Other	15	90.3 %	75.4 %	100 %
Native language				
English or bilingual	271	81.5 %	76.9 %	86.1 %
Non-English	9	69.4 %	39.3 %	99.5 %
Health literacy score				
1	136	83.2 %	76.9 %	89.5 %
2	83	83.6 %	75.7 %	91.6 %
3	47	72.9 %	60.2 %	85.6 %
4	12	70.8 %	45.1 %	96.6 %
5	2	90.0 %	48.4 %	100 %

*Percentage of MTurk workers responding that the definitions were “completely understandable” or “understandable”.

and for correlation among observations within clusters [46]. The model used a logit link and exchangeable (i.e., compound symmetry) correlation structure, since we assumed that all the variances and the covariances were equal.

To check for the possibility of interactions, we created models with interactions between the predictor variables. We compared their goodness of fit to that of the no-interactions specification using the quasi-information criterion (QIC).

Results

Table 1 shows the characteristics of the MTurk workers. The sample was 54 % female and 45 % male. With oversampling, 33 % of the MTurk workers had a high school diploma or less education, 34 % had a bachelor's or associate's degree, and 33 % had a master's degree or higher. With oversampling, 51 % of the sample were white, 23 % were Asian, 16 % were Black, 5.7 % were Hispanic, and 5.0 % were other race/ethnicity. The native language of the MTurk workers was overwhelmingly English or bilingual (97 %), while only 3.2 % were non-native English speakers. The MTurk workers on average had high health literacy: 79 % responded that they rarely or never ask their physician or pharmacist for help reading instructions.

Table 2 shows the percentages of MTurk workers responding that the definitions were “completely understandable” or “understandable” (we refer to this combination of responses as *understandable*). Overall, 81.1 % (95 % CI: 76.5–85.7 %) of the MTurk workers found the definitions understandable. 82.4 % (95 % CI: 76.3–88.5 %) of the females and 79.4 % (95 % CI: 72.4–86.5 %) of the males found the definitions understandable. Among the different age groups, understandability ranged from 76.8 % (95 % CI: 56.7 %–96.8 %) to 90.0 % (95 % CI: 56.1 %–100 %).

Among the three educational levels, perceived understandability ranged from 79.3 (95 % CI: 71.2 %–87.5 %) to 84.0 % (95 % CI: 76.5 %–91.5 %). Among the five race/ethnic groups, perceived understandability ranged from 79.2 (95 % CI: 67.2 %–91.2 %) to 90.3 % (95 % CI: 75.4 %–100 %). 81.5 % (95 % CI: 76.9 %–86.1 %) of English or bilingual speakers found the definitions understandable, while 69.4 % (95 % CI: 39.3 %–99.5 %) of non-native English speakers found the definitions understandable. Among the different health literacy levels, perceived understandability ranged from 70.8 (95 % CI: 45.1 %–96.6 %) to 90.0 % (95 % CI: 48.4 %–100 %).

Table 3 and Fig. 1 show the relationship between perceived understandability and each MTurk worker characteristic while adjusting for the other characteristics. Males were less likely to report understanding the definitions than females (OR: 0.73, 95 % CI: 0.63–0.84, $p < 0.001$). There was no clear relationship between age and perceived understandability, though the 25-to-34-year-old group was more likely than the 18-to-24-year-old group to report understanding the definitions (OR: 1.54, 95 % CI: 1.12–2.11, $p = 0.007$). There was no clear relationship between education and perceived understandability.

Asians, Hispanics, and those who marked their race/ethnicity as “other” were more likely to report understanding the definitions than whites (Asians: OR: 1.43, 95 % CI: 1.17–1.73, $p < 0.001$; Hispanics: OR: 1.86, 95 % CI: 1.33–2.59, $p < 0.001$; Other: OR: 2.48, 95 % CI: 1.65–3.74, $p < 0.001$).

Individuals whose native language was not English were less likely to report understanding the definitions than individuals whose native language was English or bilingual (OR: 0.51, 95 % CI: 0.36–0.74, $p < 0.001$).

Individuals who reported sometimes or often needing someone's help when reading instructions given by their doctor or pharmacist were less likely to report understanding definitions than individuals who reported never needing someone's help (sometimes: OR: 0.51, 95 % CI: 0.43–0.62, $p < 0.001$; often: OR: 0.40, 95 % CI: 0.29–0.55, $p < 0.001$). Individuals who reported always needing someone's help were not less likely to report understanding definitions; however, the sample size in that group was very small ($n = 2$).

There were some significant associations between perceived understandability and the definition writer. In particular, perceived understandability of definitions written by definition writers 3 and 4 were significantly lower than the reference writer ($p = 0.001$ and $p = 0.003$, respectively). Differences between the other definition writers and the reference writer were not statistically significant.

Interactions

The QIC of the GEE model we fit was 5282. This was compared with the QICs of two additional models that included the interaction between race/ethnicity and education (QIC = 5279), interaction between native language and education (QIC = 5259), and both interactions (QIC = 5254). Other combinations of interacting MTurk worker characteristics were explored but there was not enough data to fit a model. As the QIC values were almost identical, the models with interaction variables performed similarly at fitting the data. However, the sample sizes of the interaction subgroups had very low numbers of individuals, making it difficult to draw inferences (data not shown).

Conclusion

Discussion

We found that, overall, 81.1 % of the MTurk workers reported that the definitions were understandable. Males were less likely to report understanding the definitions than females. Asians, Hispanics, and those who marked their race/ethnicity as “other” were more likely to report understanding the definitions than whites. And individuals whose native language was not English and those with lower health literacy were less

Table 3
Association of MTurk workers characteristics with understandability (GEE analysis).

Predictor	n	Odds ratio	95 % CI	p
Intercept		5.24	3.60 – 7.64	< 0.001
Gender				
Female	152	Reference group		
Male	127	0.73	0.63 – 0.84	<0.001
Other	1	2.87	0.38 – 21.73	0.307
Age				
18–24	17	Reference group		
25–34	77	1.54	1.12 – 2.11	0.007
35–44	62	1.04	0.76 – 1.43	0.805
45–54	31	1.24	0.86 – 1.79	0.250
55–64	67	1.05	0.76 – 1.45	0.772
65–74	23	0.92	0.64 – 1.33	0.669
75 or older	3	2.49	1.01 – 6.16	0.048
Education				
High school diploma or less	93	Reference group		
Bachelor's or associate's degree	95	0.95	0.80 – 1.13	0.576
Master's degree or above	92	1.17	0.97 – 1.41	0.093
Race/ethnicity				
White	142	Reference group		
Asian	63	1.43	1.17 – 1.73	< 0.001
Black	44	0.96	0.78 – 1.18	0.701
Hispanic	16	1.86	1.33 – 2.59	< 0.001
Other	15	2.48	1.65 – 3.74	< 0.001
Native language				
English or bilingual	271	Reference group		
Non-English	9	0.51	0.36 – 0.74	< 0.001
Health literacy score*				
1 - Never	136	Reference group		
2 - Rarely	83	0.98	0.83 – 1.16	0.844
3 - Sometimes	47	0.51	0.43 – 0.62	< 0.001
4 - Often	12	0.40	0.29 – 0.55	< 0.001
5 - Always	2	1.66	0.59 – 4.72	0.338
Definition writer				
Definition writer 1	50	Reference group		
Definition writer 2	50	0.94	0.74 – 1.21	0.641
Definition writer 3	50	0.68	0.54 – 0.86	0.001
Definition writer 4	50	0.70	0.55 – 0.89	0.003
Definition writer 5	50	0.96	0.75 – 1.23	0.745
Definition writer 6	50	0.85	0.66 – 1.08	0.179

*Response to the question: "How often do you need someone's help when reading instructions given by your doctor or pharmacist?".

likely to report understanding definitions. We found that there were statistically significant race/ethnic differences in perceived understandability even after controlling for age, sex, education, native language, and health literacy. There was no significant relationship between education and perceived understandability. As the definitions were written for individuals with 4th-to-7th-grade reading levels, we didn't expect higher education to have a substantial effect on understanding. There were indications that higher health literacy was associated with perceived understandability, as would be expected.

Prior work has been mixed with regards to the effect of demographic information on EHR note comprehension [47]; however, across demographic groups, NoteAid has been shown to improve comprehension [22]. More generally, there is a need for health literacy materials that are written so that patients across levels of health literacy can understand them [48]. Although there are significant differences across demographic groups, overall perceived understandability of the NoteAid definitions is high. These differences are in line with prior work studying disparities in overall health literacy across demographic groups (see, e. g., [49,50]). Our results show that such high understandability can facilitate patient understanding as part of the broader NoteAid system.

We also found significant differences in perceived understandability when looking at the definition writers. We controlled for definition writer for two reasons. First, to ensure that understandability was not significantly affected by the individual writing the definitions. Second, to determine the extent to which less understandable definitions could arise given the NoteAid dictionary definition writing procedures and

potentially update the guidelines to reduce this occurrence.

Content familiarity could also affect understandability. We excluded healthcare professionals from the study. However, there are many other ways that a layperson can acquire familiarity with medical jargon, such as direct experience with health issues for themselves or family members or an interest in learning about health and illness.

It is unclear a priori what a "good" understandability level should be, given the variety in users. Prior work has shown that the related concept of familiarity ranges from 76 % to 92 % [51]. Therefore, understandability ranges in the 80–90 % range appear appropriate.

Definitions were often not complete sentences in that they might not contain a subject-verb pair. Since the Flesch-Kincaid metric was developed and validated on prose text and full sentences, this could result in a lower FKGL score. Prior work on such "imperfect" texts (e.g., blog posts, social media, etc.) have used FKGL to measure readability [32–34]. Still, FKGL is not without limitations [52,53] and so a definition scoring above 7 could be used at the definition writer's discretion.

Prior work has studied the effect of NoteAid on patient HER note comprehension; across demographic groups, NoteAid is effective in improving EHR note comprehension [21,22]. However, prior work has not considered the inherent understandability of the definitions themselves. In this work, we explicitly focus on understandability as its own upstream construct that affects downstream performance. Although there are variations in perceived understandability of definitions across groups, this does not negate the effectiveness of NoteAid.

Strengths and limitations

One of the strengths of this study is its use of a random sample of NoteAid definitions. Another strength is its oversampling of underrepresented groups. This study provided valuable feedback to the NoteAid definition writers. Because this was an observational study, we needed to control for potential confounding variables. We addressed this issue by fitting a GEE model that took into account interactions as well as main effects. Finally, we attempted to maximize the quality of the data by only using MTurk workers with high approval rating and including attention checks throughout the study instrument. Therefore, we expect our results to be representative of the larger pool of definitions in the NoteAid dictionary, further validating the benefits and impact of the overall NoteAid tool.

The definitions in NoteAid can subsequently be edited by another definition writer to correct mistakes, fix typos, grammatical errors, and other issues. In this work we only considered definitions that had not been revised by a subsequent definition writer, to make attribution to a single definition writer unambiguous, which was necessary for comparing consistency across definition writers. However, in analyzing only unrevised definitions, our results could underestimate the perceived understandability of definitions in the NoteAid dictionary.

To simplify the assessment of understandability, we excluded definitions that themselves contained jargon terms. It is unclear how this may have affected understandability ratings. Definitions with embedded jargon are allowed in the NoteAid dictionary if they substantially shorten or simplify the definition. However, they also generate a risk of further confusion, either in understanding the new term and its definition or in the complexity of navigating the tool to see both definitions. On the other hand, overly long sentences to avoid jargon use are in their own ways difficult to understand. Therefore, understandability in our sample may be higher than in the full database. Future work on this trade-off is needed.

Further, NoteAid is intended to be used to help patients understand jargon in their EHRs. However, MTurk workers are not representative of the general population. For example, they differ in age, sex, income, race/ethnicity, education, and health status [21,42,43]. As such, they may not accurately reflect understandability of NoteAid definitions among patients.

The analysis was also limited by small sample sizes in some

subgroups and modeled interactions. With a larger sample size we could have explored other interactions. For example, it could be that being a non-native English speaker has a greater effect on perceived understanding for those with higher education, because it may be a marker for English proficiency. However, our sample size for the non-English group was small. Our result is consistent with prior work [21,22], which indicates its generalizability, but more targeted work with larger samples of non-English speakers is needed.

We use the Single Item Test of health literacy, which is not as detailed as other instruments. However, it has been widely studied and has been shown to correlate with other instruments of health literacy [44,54]. What's more, its brevity and validity make it a desirable option for assessing health literacy in an online population or other time-sensitive cases such as during an already busy visit with a primary care physician. While the test was designed to distinguish between adequate and inadequate literacy, we chose to analyze each response individually. The lower health literacy groups (HL3 and HL4) were each significantly different from the reference health literacy group ($p < 0.001$) and combining them with the lowest health literacy group (HL5, $n = 2$) is unlikely to meaningfully change the significance, though it would reduce resolution.

Future work

Our results suggest that variation across definition writers accounts for some degree of reported understandability. Future work could investigate standardizing the definition writing process or incorporating formal review mechanisms into definition writing. For all of our results, replications in local patient populations would add further validity to the findings.

Future work could also take the form of focus groups to ascertain reasons for the differences in understandability by demographic characteristics that we observed. They could also elicit information about how to increase understandability.

In another study we assess the clinical accuracy of the NoteAid definitions, as well as the process of identifying jargon.

Improving understandability runs in parallel with improving patient health literacy. Even if definitions are written at a particular grade level, there are unique characteristics of medical text that make health literacy difficult, such as complex multisyllabic medical terminology and visual aids included with the text [55]. Therefore, parallel to the work being done with the NoteAid project, continued effort to increase patient access to educational materials must be done [48,56,57]. This includes the development of health education materials using resources such as the AHRQ Health Literacy Universal Precautions Toolkit [58].

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Author statements

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Declaration of competing interest

None declared.

Supplementary materials

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